

DIGITAL SIGNAL PROCESSING

Introduction to DSP Architecture and Features

An easy walk-through of how a DSP chip is built and what makes it fast at handling signals

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What is a DSP?

Definition

A Digital Signal Processor (DSP) is a specialised microprocessor designed to process digital signals — audio, video, radar, sensor data — in real time, using fast, repetitive mathematical operations.

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- Built to run the same instruction (like multiply-add) millions of times per second
 - Used inside phones, hearing aids, radios, and audio/video equipment
 - Optimised for speed and predictability, not general-purpose computing

General CPU vs. DSP

Task variety

CPU: Broad, general tasks *DSP*: Narrow: math on signals

Speed at math

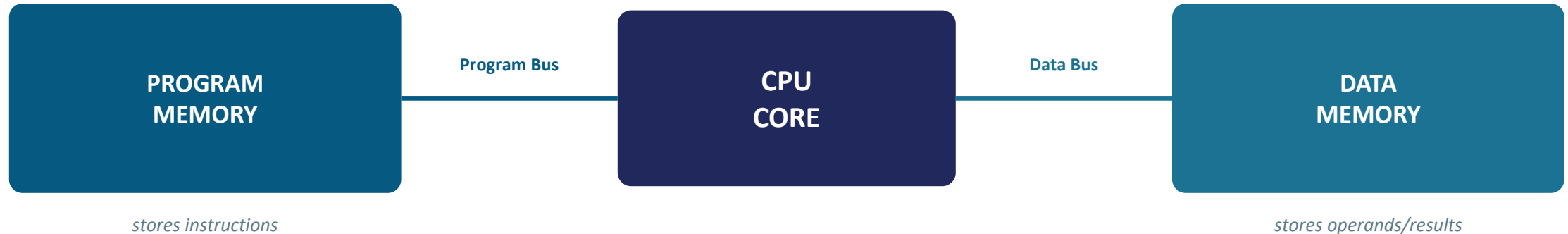
CPU: Moderate *DSP*: Very high (MAC/cycle)

Predictability

CPU: Can vary (caching etc.) *DSP*: Highly deterministic

The Harvard Architecture

Why DSPs keep program and data memory separate



In Von Neumann designs one shared bus fetches both instructions and data, creating a bottleneck. The Harvard architecture gives each its own memory and bus, so the CPU can fetch an instruction and read/write data in the same clock cycle.

Result: parallel fetch of instruction + data → higher throughput, ideal for repetitive signal math

Key Architectural Components

1

MAC Unit

Multiply-Accumulate unit performs a multiply and an add in ONE clock cycle — the core operation behind filtering.

2

ALU

Arithmetic Logic Unit handles addition, subtraction, shifting, and logical operations on data.

3

Registers

Small, ultra-fast storage that holds operands and results right next to the processing units.

4

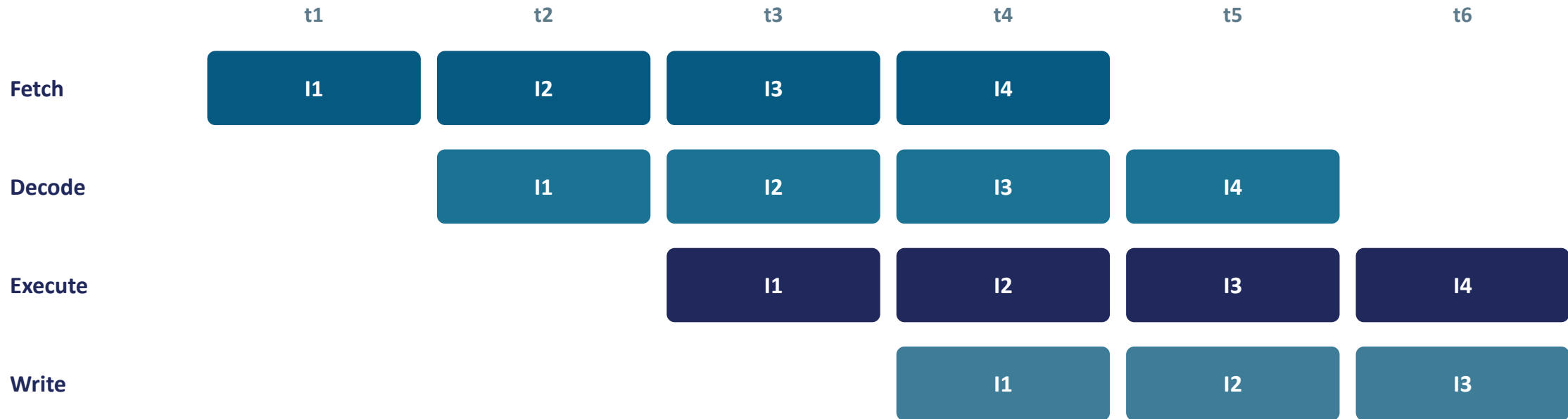
Address Generators

Dedicated units that calculate memory addresses in parallel, freeing the ALU to keep computing.

Together, **these units let a DSP repeat “multiply, add, fetch next” every single cycle** — the pattern behind digital filters and transforms.

Pipelining

Overlapping instruction stages so more work finishes per second



Without pipelining, each instruction must fully finish before the next starts. With pipelining, while one instruction executes, the next is already being decoded, and the one after that is being fetched — four instructions in flight at once.

Effect: throughput approaches one completed instruction per clock cycle, even though each instruction still takes several cycles to finish.

On-Chip Memory & Peripherals

Memory Features

On-chip RAM/ROM

Fast local memory avoids slow trips to external chips

Multiple memory banks

Lets the CPU access two operands in one cycle

DMA controller

Moves data in the background without stalling the CPU

Peripherals & I/O

Serial ports

High-speed links to sensors, codecs, other chips

Timers/counters

Generate precise timing for sampling signals

ADC/DAC interfaces

Bridge the real analog world and digital math

Why it matters: keeping memory and I/O close to the core — instead of relying only on external hardware — keeps signal-processing pipelines fast and predictable in real time.

Real-World Applications

1

Audio

Noise cancellation, equalisers, music streaming compression

2

Mobile & Telecom

Voice encoding, echo cancellation, 4G/5G signal processing

3

Medical

ECG/EEG monitors, hearing aids, ultrasound imaging

4

Automotive & Radar

Adaptive cruise control, collision detection, sonar

5

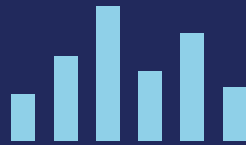
Video & Imaging

Video compression, camera image enhancement

6

Industrial

Motor control, vibration analysis, sensor fusion



Thank You

Questions are welcome

Member 1 • Member 2

